Lecture: Application of SRDE to stochastic filtering

Consider the SDE

$$(S) \quad dX_t = B(X_t)\,dt + C(X_t)\,dW_t \quad \in \mathbb{R}^d$$

under global Lipschitz conditions on B, C.
Suppose that (S) is not observed directly, but information on (S) is provided by a second process, called the observation, given as

$$(O) \quad dY_t = H(Y_t)\,dt + \Gamma\,dV_t \quad \in \mathbb{R}^p, \; Y_0 = 0$$

where H is also globally Lipschitz, (V_t) p-dim. BM independent of (W_t), and $R := \Gamma\Gamma^T > 0$.

Let $\mathcal{Y}_{0:t} := \sigma(Y_s : s \le t)$ be the σ-algebra generated by the observation process up to time t. The (optimal) filtering problem now consists of computing resp. approximating the conditional distribution

$$\eta_t(A) := P(X_t \in A \mid \mathcal{Y}_{0:t}), \quad A \in \mathcal{B}(\mathbb{R}^d).$$

Kallianpur-Striebel formula

The stochastic integral $\int_0^t \langle R^{-1}H(X_s), dY_s\rangle$ is well defined, since

$$\mathbb{E}\left(\int_0^T \|X_s\|^2 ds\right) < \infty, \text{ hence } \mathbb{E}\left(\int_0^T \langle R^{-1}H(X_s), H(X_s)\rangle ds\right) < \infty.$$

Let $Z_t(X,Y) := \exp\left(\int_0^t \langle R^{-1}H(X_s), dY_s\rangle - \frac{1}{2}\int_0^t \langle R^{-1}H(X_s), H(X_s)\rangle ds\right)$.

<u>Theorem</u>: Let $(\bar{X}_t)$ be a copy of (X_t), such that $(\bar{X}_t)$ is independent of (X,V). Then

$$\eta_t(A) = \mathbb{E}(1_A(X_t) \mid Y_{0:t}) = \frac{\mathbb{E}_{\bar{X}}(1_A(\bar{X}_t) Z_t(\bar{X},Y))}{\mathbb{E}_{\bar{X}}(Z_t(\bar{X},Y))},$$

and $\mathbb{E}_{\bar{X}}$ indicates integration w.r.t. $\bar{X}_{0:t}$ only.

<u>Proof</u>: Fix t, let $\mathcal{F}_t^{X,V} = \sigma(X_s, V_s : s \le t)$, let $P_{(X,Y)}$ (resp. $P_{(X,V)}$) the joint distribution of $(X_{0:t}, Y_{0:t})$ (resp. $(X_{0:t}, V_{0:t})$).

Then

$$P_{(X,Y)} \ll P_{(X,V)} = P_X \otimes P_V$$

with Radon-Nikodym density $Z_t(X,Y)$, since the conditional distribution $P_{Y|X}$ of $Y_{0:t}$ given $X_{0:t}$ is a Brownian motion with variance $t \cdot R$ and drift $H(X_t)$, i.e.

$$\frac{dP_{Y|X}}{dP_V} = \underbrace{\exp\left(\int_0^t \langle R^{-1}H(X_s), dV_s\rangle - \frac{1}{2}\int_0^t \langle R^{-1}H(X_s), H(X_s)\rangle ds\right)}_{= Z_t(X,V)}$$

Hence, $P_{Y|X} = Z_t(X,V) P_V$. Note that $P_{Y|X}$ is a stochastic kernel from $(C([0,t];\mathbb{R}^d), \mathcal{F}_t^X)$ to $(C([0,t];\mathbb{R}^p), \mathcal{F}_t^Y)$, hence

$$P_{(X,Y)} \underset{\text{Fubini}}{=} P_X P_{Y|X} = Z_t(X,V)\, P_X \otimes P_V \blacksquare$$

It suffices now to prove that

$$E\left(1_A(X_t)\,H(Y_{0:t})\right) = E\left(\frac{E\left(1_A(\bar{X}_t)\,Z_t(\bar{X},Y)\right)}{E\left(Z_t(\bar{X},Y)\right)}\,H(Y_{0:t})\right).$$

This follows

- from $E\left(1_A(X_t)\,H(Y_{0:t})\right) = E\left(1_A(X_t)\,H(V_{0:t})\,Z_t(X,V)\right)$

$$= \iint \underbrace{1_A(X_t)\,Z_t(X,V)\,dP_X}_{= E_{\bar{X}}\left(1_A(\bar{X}_t)\,Z_t(\bar{X},V)\right)}\,H(V_{0:t})\,dP_V$$

$$= \int E_{\bar{X}}\left(1_A(\bar{X}_t)\,Z_t(\bar{X},V)\right)\frac{E_X\left(Z_t(X,V)\right)}{E_{\bar{X}}\left(Z_t(\bar{X},V)\right)}\,H(V_{0:t})\,dP_V$$

$$= \iint \frac{E_{\bar{X}}\left(1_A(\bar{X}_t)\,Z_t(\bar{X},V)\right)}{E_{\bar{X}}\left(Z_t(\bar{X},V)\right)}\,Z_t(X,V)\,H(V_{0:t})\,dP_X\,dP_V$$

$$= E\left(\frac{E_{\bar{X}}\left(1_A(\bar{X}_t)\,Z_t(\bar{X},Y)\right)}{E_{\bar{X}}\left(Z_t(\bar{X},Y)\right)}\,H(Y_{0:t})\right) \blacksquare$$

The Zakai equation

We can use the Kallianpur-Striebel formula to obtain an SPDE for the unnormalized conditional distribution

$$\tilde{\eta}_t(A) := E_{\bar{X}}\left(1_A(\bar{X}_t)\, Z_t(\bar{X}, Y)\right)$$

as follows: fix a smooth test-function f on $\mathbb{R}^d$. Then Itô's formula implies:

$$d\left(f(\bar{X}_t) Z_t(\bar{X}, Y)\right) = Z_t(\bar{X}, Y)\, df(\bar{X}_t) + f(\bar{X}_t)\, dZ_t(\bar{X}, Y)$$

$$+ \underbrace{d\langle f(\bar{X}_\cdot), Z_\cdot(\bar{X}, Y)\rangle}_{=0 \text{ since } (V_t) \text{ indep. of } (W_t)}$$

$$= Z_t(\bar{X}, Y)\, Lf(\bar{X}_t)\, dt + Z_t(\bar{X}, Y)\, f'(\bar{X}_t)\, C(\bar{X}_t)\, dW_t$$

Itô-decomposition with
$Lf(x) = \frac{1}{2} \operatorname{tr}\left(CC^T(x) f''(x)\right) + B(x) f'(x)$
generator of (S)

$$+ f(\bar{X}_t) \langle R^{-1} H(\bar{X}_t), dY_t\rangle Z_t(\bar{X}, Y)$$

Taking expection w.r.t. $\bar{X}$ only yields

$$d\, \mathbb{E}_{\bar{X}}\left(f(\bar{X}_t) Z_t(\bar{X}, Y)\right) = \mathbb{E}_{\bar{X}}\left(\left(Lf(\bar{X}_t)\, dt + f(\bar{X}_t) \langle R^{-1} H(\bar{X}_t), dY_t\rangle\right) Z_t(\bar{X}, Y)\right)$$

and with the notation

$$\tilde{\eta}_t(f) := \mathbb{E}_{\bar{X}}\left(f(\bar{X}_t)\, Z_t(\bar{X}, Y)\right)$$

we arrive at

$$d\tilde{\eta}_t(f) = \tilde{\eta}_t(Lf)\,dt + \tilde{\eta}_t(f\cdot H)R^{-1}dY_t .$$

This is the so called <u>Zakai-equation</u> in analytically weak form. Provided $\tilde{\eta}_t(dx) = \tilde{\eta}_t(x)\,dx$ admits a smooth density we arrive at the SPDE

$$(1) \quad d\tilde{\eta}_t(x) = L^*\tilde{\eta}_t(x)\,dt + \tilde{\eta}_t(x)H(x)R^{-1}dY_t$$

where $L^*f(x) = \frac{1}{2}\sum_{i,j=1}^{d} \partial_{x_i x_j}\left((CC^T)_{ij}(x)f(x)\right) - \left(\sum_{i=1}^{d}\partial_{x_i}B_i(x)\right)f(x)$

and this is a true SPDE, driven by finite dimensional noise.

<u>The Kushner-Stratonovich equation</u>

... is the counterpart of the Zakai equation for the (normalized) conditional distribution

$$\eta_t = \frac{\tilde{\eta}_t}{\tilde{\eta}_t(1)} .$$

From the semimartingale decomposition of $\tilde{\eta}_t(f)$ we can get the —"— —"— of $\eta_t(f) = h(\tilde{\eta}_t(f), \tilde{\eta}_t(1))$ with $h(x,y) = \frac{x}{y}$.

Again Itô's formula implies:

$$d\eta_t(f) = h_x(\tilde{\eta}_t(f), \tilde{\eta}_t(1))\, d\tilde{\eta}_t(f) + h_y(\tilde{\eta}_t(f), \tilde{\eta}_t(1))\, d\tilde{\eta}_t(1)$$

$$+ \frac{1}{2} h_{xx}(\cdot)\, d\langle \tilde{\eta}_\cdot(f)\rangle + h_{xy}(\cdot)\, d\langle \tilde{\eta}_\cdot(f), \tilde{\eta}_\cdot(1)\rangle$$

$$+ \frac{1}{2} h_{yy}(\cdot)\, d\langle \tilde{\eta}_t(1)\rangle$$

$$= \frac{1}{\tilde{\eta}_t(1)}\left(\tilde{\eta}_t(Lf)\,dt + \tilde{\eta}_t(fH) R^{-1} dY_t\right) - \frac{\tilde{\eta}_t(f)}{\tilde{\eta}_t(1)^2}\, \tilde{\eta}_t(H) R^{-1} dY_t$$

$$- \frac{1}{\tilde{\eta}_t(1)^2}\, \tilde{\eta}_t(fH) R^{-1} \tilde{\eta}_t(H)\, dt + \frac{\tilde{\eta}_t(f)}{\tilde{\eta}_t(1)^3} \langle \tilde{\eta}_t(H), R^{-1} \tilde{\eta}_t(H)\rangle dt$$

$$= \eta_t(Lf)\,dt + \eta_t(fH) R^{-1} dY_t - \eta_t(f)\eta_t(H) R^{-1} dY_t$$

$$- \eta_t(fH) R^{-1} \eta_t(H)\, dt + \eta_t(f) \langle \eta_t(H), R^{-1} \eta_t(H)\rangle dt$$

$$= \eta_t(Lf)\,dt + [\eta_t(fH) - \eta_t(f)\eta_t(H)] R^{-1} dY_t$$

$$- [\eta_t(fH) - \eta_t(f)\eta_t(H)] R^{-1} \eta_t(H)\, dt$$

$$= \underbrace{\eta_t(Lf)\,dt}_{\text{Fokker Planck term.}} + \underbrace{\mathrm{cov}_{\eta_t}(f, H) R^{-1} (dY_t - \eta_t(H)\, dt)}_{\text{innovation term}} \blacksquare$$

Again, if $\eta_t(dx) = \eta_t(x)\, dx$ admits a smooth density, we arrive at the SPDE

$$\partial_t \eta_t(x) = L^* \eta_t(x)\, dt + (H(x) - \eta_t(H))\, \eta_t(x) R^{-1} (dY_t - \eta_t(H)\, dt) \blacksquare$$